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ORIGINAL RESEARCH

The use of multi-criteria decision-making methods in project portfolio selection: a literature review and future research directions

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Abstract
In most project portfolio selection (PPS) situations, the presence of multiple attributes and decision-maker preference is inevitable. As Multi-criteria Decision Analysis (MCDA) methods provide a framework well-suited to deal with these challenges in PPS problems, the use of MCDA methods in real-life PPS problems has increased in recent years. This paper provides a comprehensive literature review of the use of different MCDA methods and their individual or combined utilization with other modeling techniques to support PPS problems. First, we summarize how MCDA methods are used in different modeling approaches. Second, we examine the mathematical models that are generally used to combine MCDA with mathematical programming techniques to solve PPS problems with resource constraints. Third, we present the drawbacks of combined utilization and discuss recent advances. Finally, we visualize the summary of the reviewed papers as a decision tree to assist researchers and practitioners in the use of MCDA methods in a specific PPS context and propose some future research directions.

Keywords Multi-criteria decision analysis · Integer programming · Project portfolio selection · Preference modeling · Alternative solutions

1 Introduction

Project portfolio selection (PPS) is a complex strategic decision problem that has received increasing attention both in practice and as a field of academic research (Meskendahl, 2010; Teller & Kock, 2013; Martinsuo & Geraldi, 2020; Dixit & Tiwari, 2020). Over the years, a wide range of methods and algorithms under several approaches (i.e., financial, simulation, statistical, multi-criteria decision analysis (MCDA), mathematical programming) have been developed to deal with PPS problems (Zorluoglu & Kabak, 2021). Recently, due to the increasingly competitive and complex decision environment, such problems naturally

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often involve multiple, conflicting and incommensurate criteria (Liesiö et al., 2007; Tervonen et al., 2017; Liesiö et al., 2021). Therefore, the paradigm of MCDA provides adequate support for this type of problems (Vetschera & de Almeida, 2012; Rad & Rowzan, 2018; Demircan Keskin, 2020; Song et al., 2021; Ma et al., 2020; Nowak & Trzaskalik, 2022). Consequently, the number of publications on the use of MCDA methods for PPS has considerably increased in recent years (Morton et al., 2016).

In order to apply an appropriate MCDA method successfully for PPS, it is necessary to understand the challenges. Several studies (Archer & Ghasemzadeh, 2004; Killen et al., 2007; Müller et al., 2008; Iamratanakul et al., 2008; Frey et al., 2012; Morton et al., 2016) discuss these challenges with a special focus on the MCDA perspective. In addition to these studies, four recent literature review papers related to this topic have been published. Danesh et al. (2018) discuss the PPS decision-making challenges and provide a classification of decision-making techniques, but many critical articles that use MCDA for PPS problems are not included in this review. De Souza et al. (2021) provide a detailed review of 66 articles including the classification of both multi-objective and multi-attribute methods, analyzing the types of uncertainties and identifying the evaluation criteria. However, they focus on only the R&D portfolio problems, and therefore some major studies fall out of their scope. Furthermore, Zorluoglu and Kabak (2021) present a systematic review of 253 articles covering all kinds of decision analysis methods for PPS. They focus on a classification based on three factors: type of the study, methods used, and types of projects. As a result of a broader perspective, even though this study refers to many project portfolio studies, it lacks an in-depth analysis and provides only descriptive statistics on the classifications. Liesiö et al. (2021) discuss the recent developments and future prospects for portfolio decision analysis (PDA) in a detailed literature survey. PDA refers to a body of theory, methods, and practices that is designed for decision environments in the context of PPS with regard to multiple attribute, preferences, uncertainties and limited resources (Salo et al., 2011). In general contexts, PDA is built on MCDA to assist the decision makers in solving PPS problems. It is worth mentioning that, even though the recent PDA terminology has taken attention from some other researchers over the last years, the PPS term has been used for a few decades by a very large community. In their survey, Liesiö et al. (2021) provide an analysis of methods and models including the objectives, decision variables, constraints, uncertainties, mathematical models and solution approaches. Additionally, they apply the SWOT (strengths, weaknesses, opportunities, threats) analysis framework in order to offer directions for the further advancement of this field. However, all of these reviews do not provide information on the use of different MCDA methods; in addition, their combined utilization with other modeling techniques was neglected. Given the diversity of MCDA methods, understanding their usage in PPS would help to develop a structured framework for selecting the appropriate MCDA method/s as a tool/s in PPS decision making. This critical gap highlights the need for a new literature review.

Motivated by that gap in the literature, rather than offering a classical systematic review, our objective is to identify the different MCDA methods and to investigate how these methods have been applied in PPS problems. We also aim to compare the use of these MCDA methods, to provide a decision-tree-based summary of the reviewed papers to assist in the choice of a solution approach and to highlight future research directions. For the purposes of this study, we followed a research methodology that consists of five steps: (1) article selection, (2) classification of the selected articles, (3) review of the selected articles, (4) analysis of the mathematical formulation and (5) proposal for a selection guideline and for further research.

To select the articles, we initially used Scopus and Web of Science (WoS) as the two main databases to search for the journal articles published between January 2005 and February

2022. We then used online journal portals such as Science Direct, Springer, Taylor & Francis, World Scientific Net, Emerald and Wiley, as well as the recent literature review papers (Liesiö et al., 2021; De Souza et al., 2021; Zorluoglu & Kabak, 2021; Danesh et al., 2018) to expand the search list. Furthermore, the references of the articles were examined, as well as any other article citing these articles in order to obtain more articles that were not identified in our search. To compile the list of articles in this review, articles containing the following terms in the title, keywords or abstract were collected: "project portfolio" and "MCDA" or "MCDM" or "MADM" or "multiple criteria" or "multi-criteria" or "multicriteria" or "multiple attribute" or "multi-attribute" or "multiattribute". However, we noticed that some relevant papers did not use the word "project" in their title, abstract or keywords since a portfolio can be composed of either projects, services, products or plans depending on the context of the problem. Therefore, we removed the word "project" from the search string and performed the search again. To focus on the most relevant work, conference proceedings, book chapters and technical reports were not considered. Consequently, a total of 996 articles (628 from Scopus and 368 from WoS) were found by the search.

From those articles, we considered only non-duplicated articles in English for the following subsequent steps. Then, we applied two manual screening tasks. First, based on their abstracts, we discarded the articles discussing financial portfolio models and lacking a mathematical model for preference and/or multiple attribute modeling. It is worth noting that we excluded the studies that do not explicitly include multi-attribute evaluation of either projects or portfolios. In our second screening step, we scanned the full text of the articles. At this step, we discarded the articles that were not concerned with a generation of portfolio of alternatives even if the term "portfolio" appeared in the title, abstract or keywords. Specifically, we noticed that some articles deal only with the selection, ranking or evaluation of individual projects, and lack a model/procedure for selecting multiple alternatives to form a portfolio. In some studies, there is no additional information on how to generate a portfolio. Thus, we considered these articles as "project selection" articles and did not include them into our shortlist. In addition, we are aware of the fact that there are articles that do not have the keyword "portfolio" in their title, abstract or keywords, but involve portfolio generation. It is worth noting that we required the use of "portfolio" keyword in order to reduce the number of articles in our search. Afterwards, the articles that, upon review, proved to be beyond the defined scope were rejected. As a result, a total of 56 articles were finally selected. Despite the fact that we might have missed some potential articles due to the large volume of literature and a challenging screening process, we believe that the resulting sample of articles as the most representative ones ensures our objectives in this review study.

Each shortlisted article was studied in details. To classify articles in this review, we used the classification approach presented by Urli and Terrien (2010) and de Almeida et al. (2014a), and extended it by including the phases of the solution procedures as well as the data interaction of these phases. Then, detailed information was provided on how the MCDA methods are utilized in a combined manner with mathematical programming techniques, highlighting their advantages and disadvantages. Classifying, summarizing and reviewing the articles in this manner helped us to achieve various critical and important research insights and led us to provide recommendations for future studies.

The contribution of our review to the body of knowledge in PPS is in four ways: (1) identifying the most representative discrete MCDA articles in the literature and mapping these articles on the approach and methods, (2) investigating and comparing the approaches and methods in the field of PPS through detailed analysis of formulations and characteristics, (3) summarizing the articles using a decision tree that can support the choice of appropriate MCDA methods in a specific PPS context, and (4) highlighting areas for future research.

This paper is structured as follows: Sect. 2 provides a brief description of the PPS problem. Section 3 analyzes the modeling approaches that incorporate the use of MCDA methods in PPS problems and investigates the integrated utilization of MCDA methods with optimization techniques. Section 4 presents a decision tree based summary to assist the decision makers on how to use of MCDA methods in PPS problems. The review concludes by a discussion in Sect. 5 and highlighting promising future research directions in Sect. 6.

2 The project portfolio selection problem

Project portfolio selection (PPS) is the choice of a subset of projects (the portfolio) with the aim of maximizing the value of the portfolio with regard to multiple qualitative and quantitative criteria, subject to the given resource constraints (Morton et al., 2016). With the existence of multiple attributes and decision-maker preferences, a PPS problem is typically structured according to the (A, C, E) MCDA model. Let $A = \{A_1, A_2, \ldots, A_m\}$ be the set of candidate projects, and $C = \{C_1, C_2, \ldots, C_n\}$ be the set of criteria/attributes for evaluating the performance of projects. The evaluation e_{ij} of the ith project of A according to the jth criterion/attribute of C is represented by the multi-criteria evaluation matrix E.

According to the multi-criteria method being used, other parameters are usually required to model the decision-maker's preferences. For example, some methods introduce preferential information such as preference thresholds, preference functions or utility functions for each criterion (e.g., under the additive value function model, the overall project score is the weighted sum aggregating the value functions of each criterion). For each criterion j, the value function, denoted by v_j, ranges between 0 and 1. The multi-criteria score ms_i of alternative i is then computed as follows

$$ms_i = \sum_{j=1}^{n} w_j \cdot v_j(e_{ij}) \tag{1}$$

where w_j is the weight (relative importance) of criterion j with the weights being non-negative and normalized (w_j for all j and $\sum_{j=1}^{n} w_j = 1$).

In case of a resource constraint, the PPS problem is extended to a binary mathematical programming formulation subject to these constraints (a knapsack problem). A binary decision variable x_i equals 1 if a project x_i is selected for the portfolio and equals 0 otherwise. The selected portfolio, P, is a subset among the 2^m combinations of A ($P \in A$). Typically, that subset is defined by a resource constraint R that represents the amount of the available resource for the portfolio by r_i as the resource requirement of project A_i. From that perspective, the objective function takes on the form of maximizing the total value of selected projects for the portfolio, considering the given constraints as follows

$$\begin{aligned} &\text{maximize} \quad \sum_{i=1}^{m} ms_i \cdot x_i \\ &\text{s.t.} \\ &\qquad \sum_{i=1}^{m} x_i \cdot r_i \leq R \end{aligned} \tag{2}$$

3 Solution approaches

In recent years, MCDA was shown to provide adequate support for PPS problems given their multi-dimensional nature. A variety of methods based on MCDA techniques has been developed and applied to deal with the PPS problem in different areas (Morton et al., 2016). In general, almost all of the methods in the literature follow two main widespread alternative approaches for determining a best or optimal portfolio (Fig. 1). In the first approach (Approach-1), determining an optimal or best portfolio is done via project evaluations. Projects are evaluated individually, and then the best/optimal portfolio is generated. In the second approach (Approach-2), on the other hand, an optimal or best portfolio is achieved through portfolio evaluations. For this purpose, all possible portfolios are generated, and then these portfolios are evaluated to select the best one. A key difference between these two approaches is the decision unit. While Approach-1 focuses on the evaluation of individual projects, Approach-2 targets the evaluation of possible portfolios.

Although these approaches utilize the same methods in the decision models, solutions may differ, as the order in which the methods are applied is different. For instance, Tavana et al. (2015) first applied TOPSIS to obtain a ranking of projects, then used that ranking information as input for the objective function of an Integer Programming (IP) model to determine the optimal portfolio considering the given resource constraints. In contrast, Yang et al. (2015) first generated possible portfolios and then performed a Stochastic Multi-criteria Acceptability Analysis (SMAA) to select the best portfolio among those possible portfolios of projects.

The following sub-sections provide detailed information on these two approaches, analyzing their advantages and disadvantages. Furthermore, the methods and techniques used in these approaches are summarized in Table 1.

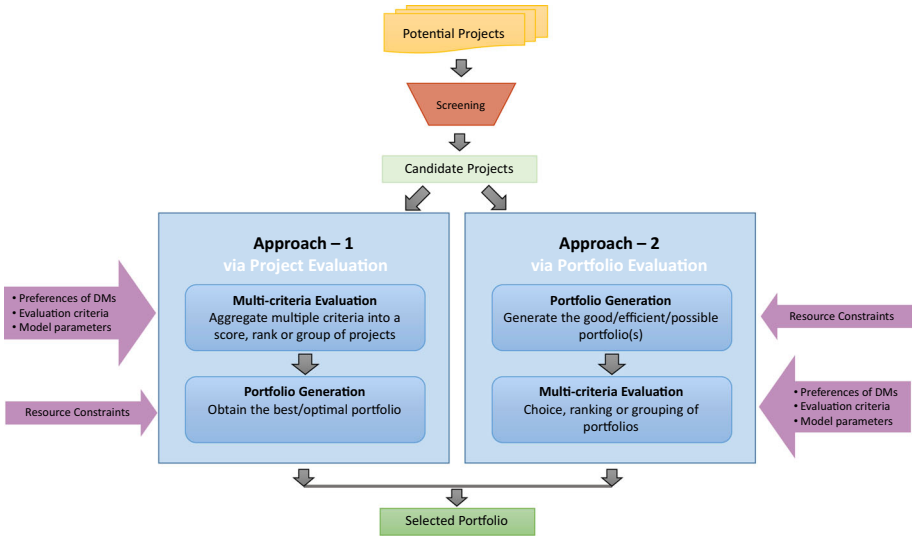

Fig. 1 Project portfolio selection approaches

Table 1 Approaches, methods and techniques for PPS

Approach	MCDA method	Optimization technique	References
Approach-1	ANP	–	Zhu et al. (2010)
	AHP & ANP	–	Aragonés-Beltrán et al. (2014)
	DEMATEL & ANP	–	Jeng and Huang (2015)
	Grey Theory—TOPSIS	–	Bhattacharyya (2015)
	DEMATEL & MABAC	–	Debnath et al. (2017)
	Fuzzy weighted average	–	Relich and Pawlewski (2017)
	AHP & TOPSIS	–	Ahriz et al. (2018)
	AHP & TOPSIS	–	Hashemizadeh and Ju (2019)
	MAVT	–	Parreiras et al. (2019)
	AHP	–	Szilágyi et al. (2020)
	YODA	–	Kurttila et al. (2020)
	Fuzzy TOPSIS	–	Ma et al. (2020)
	AHP & WASPAS	–	Rudnik et al. (2021)
	Benefit-to-cost ratio (BCR)	–	Frej et al. (2021)
	DEMATEL & Fuzzy inference system	–	Hajiagha et al. (2022)
	ANP	Goal Programming	Lee and Kim (2001)
	ELECTRE-TRI	IP	Mavrotas et al. (2003)
	PROMETHEE	IP	Mavrotas et al. (2006)
	Additive utility function	IP	Mavrotas et al. (2008)
	TOPSIS	MOLP	Zandi and Tavana (2010)
	PROMETHEE	IP	Shakhsi-Niaei et al. (2011)
	Weighted Sum	Non-linear 0-1 Programming	de Almeida and Duarte (2011)
	MAUT	MONLP	Yu et al. (2012)
	PROMETHEE	IP	Vetschera and de Almeida (2012)
	PROMETHEE	IP	de Almeida and Vetschera (2012)
	DEA & TOPSIS	Artificial bee colony (ABC) Algorithm	Chang and Lee (2012)

Table 1 continued

Approach	MCDA method	Optimization technique	References
	PROMETHEE	IP	Mavrotas and Pechak (2013b)
	PROMETHEE	IP	de Almeida et al. (2014b)
	DEA & TOPSIS	IP	Tavana et al. (2015)
	MAUT	IP	Lopes and de Almeida (2015)
	AHP	IP	Parvaneh and El-Sayegh (2016)
	Additive Utility Function	IP	Martins et al. (2017)
	Fuzzy AHP & PROMETHEE	IP (Genetic Algorithm)	Wu et al. (2018)
	DEA & Fuzzy ANP	Multi-Objective Non-linear Programming	Sharifighazvini et al. (2018)
	Fuzzy AHP & Weighted Sum	Non-linear 0-1 Programming (Genetic Algorithm)	Wu et al. (2019a)
	Fuzzy AHP	Non-linear 0-1 Programming (Genetic Algorithm)	Wu et al. (2019b)
	Fuzzy AHP	MOLP	Tavana et al. (2019)
	DEMATEL & ANP	MOLP	Raad and Shirazi (2020)
	Fuzzy ANP	MONLP	Demircan Keskin (2020)
	VIKOR	Goal Programming	Zhang et al. (2020)
	TOPSIS	IP	Tavana et al. (2020)
	AHP	MOLP	Raad et al. (2020)
	Weighted Average Method	IP	Mavrotas and Makryvelios (2021)
	MACBETH	Non-linear 0-1 Programming	Delouyi and Ashrafi (2021)
	AHP & SMART	IP	Guangul et al. (2021)
	Fuzzy AHP & TOPSIS	IP	Ranjbar et al. (2022)

Table 1 continued

Approach	MCDA method	Optimization technique	References
Approach-2	DEA	Branch and Bound	Eilat et al. (2006)
	DEA	Metaheuristic	Urli and Terrien (2010)
	DEA	Non-dominated sorting genetic algorithm II	Tavana et al. (2013)
	SMAA	Not specified	Yang et al. (2015)
	TOPSIS	Multiple objective particle swarm optimization	Rad and Rowzan (2018)
	TOPSIS	MONLP	Zhang et al. (2019)
	SMAA	Backtracking algorithm	Song et al. (2019)
	SMAA	Genetic & Simulated annealing algorithm	Song et al. (2021)
	TOPSIS & AHP	Genetic algorithm	Mussoi and Teive (2021)
	Robust ordinal priority approach	OWA and Fuzzy C-Means	Mahmoudi et al. (2022)

MOLP Multi-objective linear programming; *MONLP* Multi-objective non-linear programming

3.1 Approach-1 (project evaluation)

In the first approach, initially a multi-criteria evaluation of the individual projects is carried out using an MCDA method that accounts for both the qualitative and quantitative criteria. Morton (2015) studied the choice of measurement for project benefits in multi-criteria portfolio decision analysis by using a derived measurement frame. This study is a useful starting point to practice an MCDA method for the PPS problem, as it provides directions on how to evaluate individual project benefit scores with respect to various criteria.

Depending on the problem's characteristics and decision-makers' preferences, regardless of the number of candidate projects, various types of MCDA methods that generate different outputs can be chosen from the literature. For instance, while some authors aimed to determine a score (utility or value) for each project to clearly present the contribution of individual projects to the portfolio, some preferred methods that generate a ranking of projects showing the priority of projects in the portfolio. Furthermore, few authors used sorting methods that generate groups of projects based on their calculated preferences.

The methods used in this category include ANP (Zhu et al., 2010), AHP/ANP (Aragonés-Beltrán et al., 2014; Szilágyi et al., 2020; Rudnik et al., 2021), an integrated methodology based on DEMATEL and ANP (Jeng & Huang, 2015), Fuzzy TOPSIS (Ma et al., 2020), TOPSIS-like Grey Theory methodology (Bhattacharyya, 2015), a combined grey DEMATEL and Multi-Attributive Border Approximation Area Comparison (MABAC) (Debnath et al., 2017), Fuzzy Weighted Average (FWA) (Relich & Pawlewski, 2017), integrated AHP and TOPSIS method (Ahriz et al., 2018; Hashemizadeh & Ju, 2019), Multi-Attribute Value Theory (MAVT) (Parreiras et al., 2019) and Your Own Decision Aid (YODA) (Kurttila et al., 2020). As seen in Table 1, AHP/ANP were the most popular methods.

Based on the different MCDA outputs (i.e. values, rankings or sorting of the candidate projects), decision-makers generate a portfolio by considering other factors (i.e., available budget, cost of projects, interactions between the projects) that are not included within the MCDA model. For this purpose, one of the most common procedures is to define a cut-off position in the ranking to select the projects according to the available budget. Such a procedure ensures efficient portfolios only if the costs of projects are close to each other and the cost differences are small. Otherwise, when the costs of projects differ significantly from one another, procedures such as benefit-to-cost ratio (e.g., Frej et al., 2021) become a favorable solution instead of their pure scores/ranking. Furthermore, firstly allocating the resources across the groups/categories and subsequently across the projects belonging to each group/category is another way of creating a portfolio (e.g., Parreiras et al.,2019). In this way, it is also possible to consider each group separately by setting objectives, criteria, and a decision model for each group.

In the case of many resource-, segmentation- and other constraints, these procedures should turn into a 0-1 knapsack problem. By treating this problem with proper optimization techniques, one can obtain optimal/efficient portfolio(s). Such an approach then falls into the two-phased combined procedure that is a common way to deal with both multiple-criteria evaluation and given resource constraints in a PPS problem. In this procedure, first an MCDA method is applied to determine the value/ranking/sorting of each project, and then this information is utilized in the objective function of a mathematical programming model that is able to incorporate the resource constraints. Specifically, the value/ranking/sorting of projects that results from the first phase of the analysis is often used in an additive objective function of an optimization model in the second phase. Usually, the optimization model has a knapsack form in order to maximize the total value/ranking/sorting of selected projects for the portfolio subject to resource and/or multiple other constraints as presented in Eq. 2.

This two-phased procedure generates portfolios by combining methods such as the following: ANP and Goal Programming (GP) (Lee & Kim, 2001), VIKOR and GP (Zhang et al., 2020), ELECTRE-TRI and IP (Mavrotas et al., 2003), Additive Utility Function and IP (Mavrotas et al., 2008; Martins et al., 2017), PROMETHEE and IP (Mavrotas et al., 2006; Shakhsi-Niaei et al., 2011; Vetschera & de Almeida, 2012; de Almeida & Vetschera, 2012; Mavrotas & Pechak, 2013a; de Almeida et al., 2014b; Wu et al., 2018), MAUT and IP (Lopes & de Almeida, 2015), TOPSIS and IP (Tavana et al., 2015), AHP and IP (Parvaneh & El-Sayegh, 2016; Tavana et al., 2019), and SMAA and four heuristic algorithms (particle swarm optimization, genetic algorithm, simulated annealing algorithm and teaching-learning based optimization) (Song et al., 2021). Furthermore, Multi-Objective Linear Programming is performed with TOPSIS (Zandi & Tavana, 2010), Fuzzy AHP (Tavana et al., 2019) and DEMATEL & ANP (Raad & Shirazi, 2020). In addition to these, Non-linear Programming is also utilized with the MCDA methods such as Weighted Sum (de Almeida & Duarte, 2011; Wu et al., 2019a), TOPSIS (Chang & Lee, 2012), AHP (Wu et al., 2019a, b; Demircan Keskin, 2020) and ANP (Sharifighazvini et al., 2018). As seen in Table 1, most of the PPS methods proposed in the literature rely on the first approach, i.e., project evaluation, specifically two-phased combined procedures that use PROMETHEE/AHP and IP methods.

3.2 Approach-2 (Portfolio evaluation)

The second approach focuses on multi-criteria portfolio evaluation instead of individual project evaluation. Although this evaluation seems to be similar to the one in Approach-1, it is worth observing that in this approach, the decision unit is the portfolio, not the project. This approach initially generates all efficient portfolios by considering relevant constraints and then evaluates these portfolios to select the best one. The notion of efficient portfolio relates generally to a non-dominated portfolio, but it can also be understood as a pareto optimal, possible, maximal or feasible portfolio (Urli & Terrien, 2010). Techniques such as branch and bound algorithm (Eilat et al., 2006; Ghapanchi et al., 2012) and metaheuristics (Doerner et al., 2004; Urli & Terrien, 2010; Tavana et al., 2013; Rad & Rowzan, 2018) are utilized. In this first phase, it is important to note that there is no preference aggregation. In the second phase, an MCDA analysis is applied in order to obtain a ranking of the non-dominated portfolios or to choose the best compromise portfolio. For this purpose, Eilat et al. (2006), Urli and Terrien (2010), Ghapanchi et al. (2012) and Tavana et al. (2013) applied Data Envelopment Analysis (DEA), Rad and Rowzan (2018), Zhang et al. (2019) and Mussoi and Teive (2021) performed TOPSIS method, and Yang et al. (2015), Song et al. (2019) and Song et al. (2021) proposed to use the Stochastic Multi-attribute Acceptability Analysis (SMAA) method. In all of these studies, this approach was applied for only a small number of projects.

It is important to note that in the literature, there is also a stream of studies that has made no assumption about the decision maker's preferences to determine the optimal portfolio or the efficient/possible portfolios. These studies tend to aggregate the decision to a limited number of objective(s) and utilize an optimization model to develop an optimal portfolio directly from the candidate projects that provides maximum benefit (i.e., net present value) while dealing with project interactions, resource dependencies and constraints, or other considerations. There exists plenty of different methods and tools, which are generally based on programming methods, such as linear, nonlinear, integer, dynamic, goal, multi-objective, fuzzy, stochastic, and heuristic methods (Carazo et al., 2012; Castillo & Dorao, 2013; Jafarzadeh et al., 2015; Fernandez et al., 2015; Mashayekhi & Omrani, 2016; Dixit & Tiwari, 2020). Since in this study we focus on the use of different MCDA methods, these studies, where either DM's

Table 2 Comparison of PPS selection approaches

Approach	Advantage	Disadvantage
Approach-1	Appropriate for dealing with project-level qualitative data	Bias toward selecting the low-cost projects (due to objective function coefficient problem)
	Visibility of the impact of individual projects on overall portfolio	
Approach-2	Appropriate for dealing with synergies/interrelationships	Very large portfolio space in case of large number of projects
	Easy to understand the differences between portfolios	Computation time challenges
		Difficulty of evaluation of large number of portfolios under multiple criteria

preferences or the multi-criteria evaluation are not considered to determine a final portfolio, are beyond the scope of this review paper.

3.3 Advantages and disadvantages of the two approaches

As the first approach performs a multi-criteria evaluation of the individual projects for calculating the final score or the ranking of each project, it provides a more effective way to deal with both qualitative and quantitative criteria on the project level. Furthermore, this approach can also easily define the contribution of individual projects to the overall portfolio as the portfolio score is mostly the summation of the overall values of the projects that are included in the portfolio (Liesiö & Punkka, 2014). Because of these advantages, Approach-1 is regarded to be very promising with a wide applicability. However, these approaches also provide a major challenge as there tends to be a bias towards selecting the low-cost projects due to the objective function coefficients when mathematical programming is applied after MCDA (Mavrotas et al., 2008; Tavana et al., 2015). We will give more detailed information on this bias due to the scaling issues in knapsack formulation and procedures to overcome this drawback in a separate in Sect. 3.3.2.

The application of the second approach to the PPS problem has two main advantages: first it can handle the project interactions as it aggregates the project evaluations to generate possible portfolios, and second, it helps to provide a comprehensive evaluation of the portfolios. However, it has a few computational and measurement challenges in case of a large number of projects. We will provide detailed information on this topic and alternate ways to overcome these weaknesses in Sect. 3.3.1.

Table 2 summarizes these main advantages and disadvantages of the two approaches. As both of them have advantages and disadvantages, the selection of an appropriate approach depends on the few factors such as the availability of the resource constraint, the number of candidate projects, the output of the applied MCDA method as well as the decision-makers. However, there exists solutions to avoid the disadvantage of the Approach-1 that makes this approach more favorable than Approach-2. Next section provides these solutions in details.

3.3.1 Computational issues in portfolio generation

In practice, many organizations face PPS problems with a moderately large number of projects. In such problems, the second approach which initially generates all efficient

portfolios and then evaluates these portfolios under multiple criteria and decision-makers' preferences to select the best one, becomes very challenging in terms of computation and evaluation of non-dominated portfolios.

First, if the number of projects increase (i.e., 2437 project proposals in a case study of Mavrotas and Makryvelios 2021) and the volume defined by the given preferences becomes complex (i.e. the feasible weight space defined by the various linear constrains as in Tervonen et al. 2017), the number of non-dominated portfolios increases exponentially. For instance, in the case of over 20 projects, there may be millions of possible portfolios to evaluate. It is important to note that in the literature, there are several exact algorithms (e.g., Liesiö et al. 2008) and approximate algorithms (e.g., Mild et al. 2015; Tervonen et al. 2017) for solving medium-sized multi-objective problems within a reasonable amount of time. Besides theses studies, there are few recent studies [i.e., Marques et al. (2022), Nowak and Trzaskalik (2022), Petchrompo et al. (2022b), Petchrompo et al. (2022a)] that proposed alternate ways of generating efficient and viable portfolios in a reasonable time. Marques et al. (2022) used the concept of c-optimal portfolios (see the details of the original concept in de Almeida and Vetschera (de Almeida & Vetschera, 2012)) and refinement strategies of feasibility and efficiency during the process of portfolio generation while keeping the computational efforts within reasonable limits. Furthermore, Nowak and Trzaskalik (2022) proposed a new trade-off multi-objective dynamic programming procedure that allows the decision makers interactively apply trade-offs between the projects. Petchrompo et al. (2022b) used pruning methods to generate a subset of representative non-dominated solutions to address a multi-objective portfolio problem. Their solution ensures that the subset is selected in such a way that it strongly represents the original set of non-dominated solutions by maximising the solution diversity. Petchrompo et al. (2022a) presented a review of Pareto pruning methods for multi-objective optimization problems. However, this issue still remains a major challenge for many real-life PPS problems such as the one in Mavrotas and Makryvelios (2021).

Second, when the set of non-dominated portfolios is so large, it becomes difficult or sometimes impossible to evaluate those portfolios under multiple criteria to select the best/optimal one. In addition, using robustness measures such as maximin and minimax-regret for this purpose is not advised in this kinds of situations since they permit far less conclusive decision recommendations (Liesiö et al., 2007; Baker et al., 2020). Instead, project or portfolio level robustness robustness indices computed from the non-dominated portfolios becomes a favorable option to comprehend and make a well-informed decision (Tervonen et al., 2017; Mavrotas & Makryvelios, 2021; Kandakoglu et al., 2022). Consequently, most existing studies that use this approach analyze only a small number of portfolios (i.e. less than a hundred). For instance, Yang et al. (2015) dealt with only six projects and six possible portfolios, Eilat et al. (2006) worked with nine projects and obtained 56 feasible portfolios, Urli and Terrien (2010) generated seven portfolios out of nine projects, and finally Rad and Rowzan (2018) deals with only 10 projects. Therefore, the difficulty of building non-dominated portfolios with many projects makes this approach hard to apply to real-world problems (Urli & Terrien, 2010; Song et al., 2021).

3.3.2 Scaling issues in knapsack formulation

Generally, in PPS problems, the motivation behind using a two-phased procedure based on combined MCDA and IP methods is to consider both multiple criteria and resource constraints. Hence, most of the studies focus on two-phased procedures (Table 1) in which the objective function is defined as the sum of the multiple criteria scores of the selected project for the optimal portfolio (Eq. 2).

The drawback of the knapsack formulation is that, due to the constraints (i.e. budget) and the additive objective function of the optimization model, the ranking of the projects according to their multi-criteria scores is not respected (Mavrotas et al., 2006; de Almeida & Vetschera, 2012; Kandakoglu et al., 2022). Projects with low scores and low costs may be preferred over better projects with higher costs (Mavrotas et al., 2008). For example, assume that we have three proposals, A, B and C, with multi-criteria scores 0.75, 0.5, 0.3 and costs 100K, 40K and 50K, respectively. If we directly use the multi-criteria scores as the objective function coefficients, then the combination of projects B and C will outrank project A because B and C have aggregate score $0.5+0.3 \geq 0.75$ and aggregate cost $40K+50K \leq 100K$. The IP model will select the combination of B and C instead of A, although individually B and C have lower scores than A. This situation occurs mainly because the above-mentioned IP formulation, by its nature, allows for the comparison of a single project with a combination of projects. This is undesirable in most decision situations because the major concern is to select the best projects for the portfolio.

To overcome the above deficiency, some authors have provided different solutions presented in Table 3. Mavrotas et al. (2003) formulated the objective function as the maximization of the average instead of the sum of the scores of the projects as follows:

$$\text{maximize} \frac{\sum_{i=1}^{m} ms_i \cdot x_i}{\sum_{i=1}^{m} x_i} \tag{3}$$

The above formulation is no longer linear because the denominator of the objective function contains decision variables. Even though the authors managed to transform this non-linear model into an equivalent MILP, the new model was not applicable due to its large size and the complex structure, which resulted in a time-consuming solution procedure (Mavrotas et al., 2006). In their next study, Mavrotas et al. (2006) introduced two new ideas. First, they transformed all the net flows obtained with PROMETHEE II into non-negative scores by subtracting the most negative (lowest) net flow from the net flows of all the projects. They did that to address the negative net flows that may exist in PROMETHEE II, whereas all the coefficients in the knapsack formulation should be positive in order for all the alternatives to be eligible in the maximization objective. Second, they presented a parametric solution to solve the knapsack model (Eq. 2) by including an additional constraint as follows:

$$\sum_{i=1}^{m} x_i \leq P(k) \tag{4}$$

where $P(k)$ is the given maximum number of projects. Then, they proposed to iterate this procedure to easily detect the optimal value P^*, which is the value of $P(k)$ for which the total budget of the solution is closest to the available budget. With this formulation, the ranking of the projects according to their multi-criteria score is respected unless there are cases in which the other constraints are violated. This approach was also proved to be more effective than the maximization of the average approach (Eq. 3).

de Almeida et al. (2014b) denoted this problem as the portfolio-size effect. Similar to the parametric solution, de Almeida and Vetschera (2012) and Vetschera and de Almeida (2012) developed an alternative approach based on the concepts of c-optimal portfolios by adding the following constraint to the original problem to overcome the portfolio-size effect:

$$\sum_{i=1}^{m} x_i = c \tag{5}$$

Table 3 Two-phased combined methods for PPS

Article	MCDA method	Output	Bias solution	Opt. tech.	Objective function
Lee and Kim (2001)	ANP	Value	–	GP	$min P_k wd^- + wd^+$
Mavrotas et al. (2003)	ELECTRE-TRI	Value	Averaging scores	IP	$max \frac{\sum_{i=1}^{m} ms_i \cdot x_i}{\sum_{i=1}^{m} x_i}$
Mavrotas et al. (2006)	PROMETHEE	Value	Parametric function	IP	$max \sum_{i=1}^{m} \Phi_i + t \cdot x_i$
Mavrotas et al. (2008)	Utility function	Ranking, Class	Augmented scores	IP	$max \sum_{i=1}^{m} as_i \cdot x_i$
Shakhsi-Niaei et al. (2011)	PROMETHEE	Ranking	Augmented scores	IP	$max \sum_{i=1}^{m} as_i \cdot x_i$
Vetschera and de Almeida (2012)	PROMETHEE	Value	c-optimal	IP	$max \sum_{i=1}^{m} \Phi_i \cdot x_i$
de Almeida and Vetschera (2012)	PROMETHEE	Value	c-optimal	IP	$max \sum_{i=1}^{m} \Phi_i \cdot x_i$
Mavrotas and Pechak (2013b)	PROMETHEE	Value	Augmented scores	IP	$max \sum_{i=1}^{m} as_i \cdot x_i$
de Almeida et al. (2014b)	PROMETHEE	Value	c-optimal	IP	$max \sum_{i=1}^{m} \Phi_i \cdot x_i$
Lopes and de Almeida (2015)	MAUT	Value	–	IP	$max \sum_{i=1}^{m} ms_i \cdot x_i$
Tavana et al. (2015)	TOPSIS	Ranking	Augmented scores	IP	$max \sum_{i=1}^{m} as_i \cdot x_i$
Parvaneh and El-Sayegh (2016)	AHP	Value	–	IP	$max \sum_{i=1}^{m} ms_i \cdot x_i$
Martins et al. (2017)	PROMETHEE	Value	c-optimal	IP	$max \sum_{i=1}^{m} \Phi_i \cdot x_i$

where c is a given portfolio size. By varying c, different portfolios are obtained, which can then be compared to each other at the portfolio level using any multi-criteria decision method (de Almeida & Vetschera, 2012). The c-optimal concept was later utilized in other studies (de Almeida et al., 2014b; Martins et al., 2017).

Furthermore, Mavrotas et al. (2008) did not maximize the aggregated performance function, but maximized the compatibility of the final selection with the initial ranking of the projects. For this purpose, they used a technique to replace the original multi-criteria scores with the "augmented" scores in the objective function. Using the augmented scores, the projects obtained in the portfolio are consistent with the multi-criteria evaluation results. The augmented scores can be obtained from either multi-criteria ranking or multi-criteria sorting results. This approach allows for more flexibility in terms of using MCDA ranking and sorting methods along with value-based methods. Later, Shakhsi-Niaei et al. (2011) and Tavana et al. (2015) utilized the augmented score technique. In addition, Clemen and Smith (2009) discussed the importance of establishing an appropriate baseline score for not selecting a project in multi-attribute portfolio analysis. Later, Liesiö and Punkka (2014) developed a technique to elicit a suitable baseline value, and Morton (2015) studied the settings of baselines from a theoretical measurement point of view.

In summary, to overcome bias while using the two-phased procedure under Approach-1, different authors provided alternate solutions such as averaging scores, parametric approaches, c-optimal, augmented scores and baseline selection. However, all solutions introduce additional challenges to the initial problem, and there is no consensus on which solution is the most effective. For instance, one has to perform an additional optimization model to generate augmented scores, or one may have to apply an additional MCDA method to evaluate many alternate c-optimal portfolios that have varying number of projects. Furthermore, while augmented scores can be generated from the value, ranking or sorting output of an MCDA model, baseline and c-optimal approaches need a value output. In this sense, it is essential to figure out how one can use an MCDA method in an effective way. Thus, in the next section, we present a summary of reviewed papers to assist researchers and practitioners in the use of MCDA methods for PPS, including the bias solution approach.

4 Summary of reviewed papers using MCDA for PPS

After reviewing variants of approaches and methods, in this section we propose a decision tree based summary of reviewed papers to support the choice of an appropriate solution method for a specific PPS problem. The decision tree structured on three main levels represented in Fig. 2 has decision nodes corresponding to the main portfolio characteristics such as portfolio generation method, number of projects, existence of resource and relevant large number of constraints, and target of MCDA output. It has also branches linked by choice nodes, and leaf nodes representing the solution method to be applied. The users move through the tree by narrowing down their options by following the branches, usually responding to a specific question (e.g., do you have a large number of projects, for instance more than 100?) at each node. It has a natural "if... then... else..." structure that allows the user to check systematically to determine a suitable branch and finally a candidate method.

It is important to note that including all the different aspects and characteristics of a PPS problem into a single decision-tree is difficult. It may result in a combinatorial explosion of the nodes and branches. Therefore, in this study, similar to the rules system-based decision

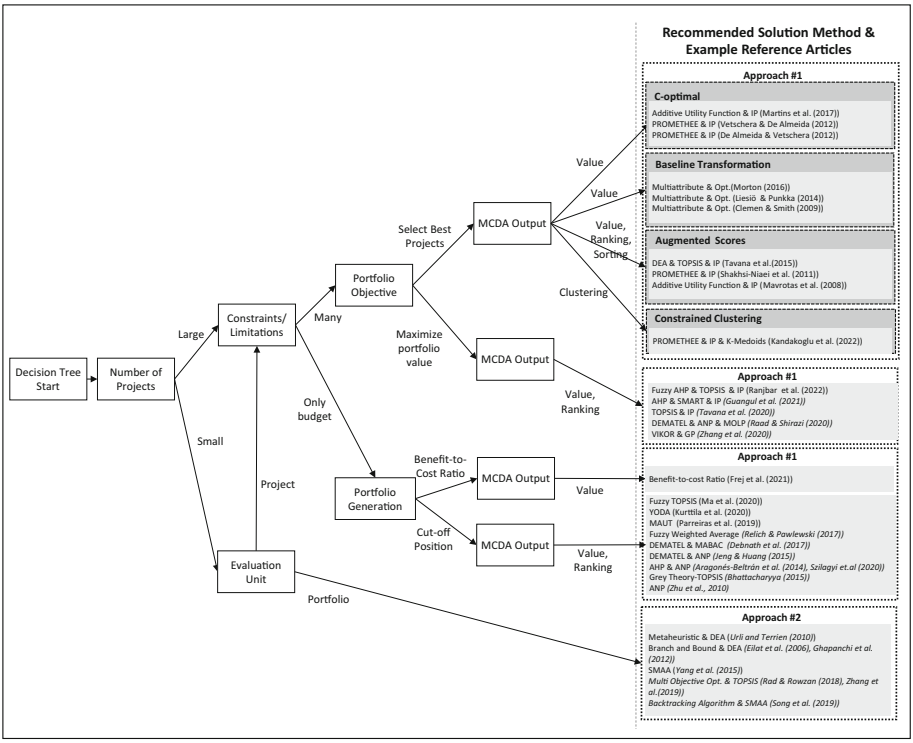

Fig. 2 Decision tree-based summary of reviewed articles

tree proposed by Cinelli et al. (2020), we prefer to include a few critical aspects in the tree structure to make it practical and easy to apply.

A limitation of the proposed decision-tree is that it may lead to the selection of a set of methods rather than a single method. In such cases, the user should select the appropriate method from the set of proposed methods. Furthermore, in some situations, the questions to make a decision are not clear to the users, specifically to the practitioners. In this case, as soon as one condition of a rule is met (e.g., number of projects), a specific method is recommended (i.e., AHP), even if there would be another that would fit as well (e.g., PROMETHEE II).

The proposed decision tree is explained using the top-down procedure in details below.

We start moving through the tree with the number of project since this is the most significant factor in the choice of an approach and a specific method. If the number of projects is small and multicriteria evaluation of portfolios are required, all possible portfolios should be generated using a mathematical programming technique by considering the relevant constraints and then an MCDA method should be utilized in order to evaluate portfolios. This includes the combined methods applied in the Approach-2.

In case of a need for a project-level multicriteria evaluation, the number of projects becomes a non-critical factor for the decision. Instead, resource and relevant constraints is a significant factor in this decision. If there is only budget constraint or there are simple constraints that do not require an additional optimization model, the choice of a suitable method depends on the portfolio generation procedure and the MCDA output. An MCDA method (e.g., AHP, ANP, MAUT, TOPSIS, PROMETHEE II etc.) that creates a value or a

ranking output in Approach-1 should be performed. At this point, users can use the decision support tool developed by Cinelli et al. (2020) to select the specific MCDA methods based on many other characteristics of a decision problem such as the structure of criteria, evaluation of performance, elicitation of preferences and features of aggregation.

For situations with complex resources and relevant constraints, one should use a two-phased combined procedure, in which, first individual project evaluations are carried out by an MCDA method and then constraints are incorporated by a mathematical programming model. In this case, the next decision node is the objective of the portfolio decision. When the user aims to maximize the total value of the portfolio (the sum of the multiple criteria scores of the selected project for the optimal portfolio), methods such as AHP, TOPSIS, PROMETHEE II followed by an IP model become a favorable option. Note that the net preference flow in PROMETHEE II and similarity (distance) in TOPSIS are considered as the value of a project. In case of a portfolio objective of selecting the best projects, the choice of a final solution method depends on the MCDA problem statement and the applicability of the methodology.

The user has to choose from among the different types of solutions: c-optimal, baseline conversion, augmented scores or constrained clustering. If the user decides to use ranking or sorting types of MCDA output, augmented scores becomes the only choice to proceed. In that method, the ranking or the sorting information is first converted into augmented scores, and then these scores are used in the objective function of the optimization phase to ensure that the optimal portfolio is consistent with the initial ranking/sorting of the projects. On the other hand, if the MCDA method generates values, then these values can be used either in c-optimal, augmented scores or baseline conversion procedures. If one is able to determine a baseline for each criterion, baseline conversion can be a viable option to implement. However, most often, it is difficult to identify a baseline, which makes this method hard to apply in practice. In such cases, c-optimal and augmented-scores become preferable alternative solutions. One should note that c-optimal procedure requires a parameter c for the optimization model to fix the number of projects in a portfolio and to generate alternate portfolios with different sizes. Then, it also requires the decision-makers to be involved again in a further step that includes the application of another MCDA model to select the best portfolio. On the other side, augmented-scores requires a further step that incorporates an algorithm to generate the scores without decision-makers involvement. At this point, the choice among these two solutions depends on the availability of the decision-makers. Finally, there is another efficient option based on a multicriteria constrained clustering methodology that enables to group the similar projects into an optimal portfolio cluster that complies with the given constraints. If the user has no preference on the MCDA output, this methodology is recommended.

It is important to note that the proposed decision-tree is developed based on the reviewed articles. Due to this fact, the user sometimes cannot find a suitable branch to move. In this kind of situation, we recommend the user selects an approach and a method by considering the advantages and disadvantages of the approach and methods.

Once a decision tree has been constructed, it is more usable to convert it into an equivalent set of rules. Decision rules are more understandable by the decision makers since their IF-THEN structure semantically resembles natural language and the way we think. Below are the example rules derived from the decision tree in Fig. 2:

– IF number of projects = small AND multi-criteria evaluation unit = portfolio, THEN solution method = Approach-2.

– IF number of projects = large AND resource constraint = many AND portfolio objective = select best projects AND MCDA output = clustering, THEN solution method = Approach-1 using Multi-criteria Constrained Clustering.
– IF number of projects = large AND resource constraint = only budget AND portfolio generation = cut-off position), THEN solution method = Approach-1 using any of the ranking methods.

The above rules can be conveniently applied by PPS practitioners and/or scholars to determine an appropriate method for solving problems according to the specific characteristics of those problems.

5 Discussion and recommendations for future research

Although multiple criteria and multiple preferences of decision makers are common and typical in real-life PPS problems, they are not considered in most case-studies. As demonstrated by the articles reviewed, MCDA methods showed a great potential to be an effective evaluation tool in dealing with preference assessments in PPS problems. The use of MCDA methods to support the PPS process generates a wide range of questions with practical implications for researchers and practitioners. First, we discovered that the stand-alone utilization of MCDA methods in PPS problems is limited to a few papers. Our review shows that the authors focused on using MCDA methods in combination with optimization methods to incorporate resources and other constraints. The PROMETHEE-IP and AHP-IP integrated methods are observed to be the most widely used methods. From this point of view, we can conclude that MCDA methods are utilized as a supporting tool for optimization techniques. Second, while there are many possible MCDA methods that can be used in selecting a portfolio, there is no consensus on which ones are the most effective. Each organization tends to choose one or more methodologies that suit their organizational culture and that make it possible to consider the most important project attributes for the project sets being considered (Archer & Ghasemzadeh, 2004, 2007).

In conclusion, the following future research directions can be suggested based on our literature review analysis:

Combined utilization with optimization techniques: This review shows that MCDA methods were mostly utilized in an integrated manner with optimization models to deal also with several constraints. The two-phased combined procedures aim to maximize the overall value of the projects that are included in a selected portfolio. Thus, the value of a portfolio becomes an aggregate of the values of the individual projects.

As mentioned in Sect. 3.3.2, the two-phased procedures cause a bias towards low cost projects to be included in the portfolio. However, to overcome this problem, some authors utilized transformations such as using augmented scores, creating c-optimal portfolios and choosing a baseline. There is no consensus on which one is the most effective way to solve this problem. Therefore, future work should develop new theoretical frameworks or extend the current ones to eliminate the scaling problem.

Objective of portfolio decisions: This review highlights that while most PPS applications in the literature assume that the portfolios with higher values are preferred by the decision makers, this objective is not valid in some situations. Although there is a necessity to replace this common objective function, the ones presented in this study (i.e., Mavrotas et al., 2008; Shakhsi-Niaei et al., 2011; de Almeida and Vetschera, 2012; Morton, 2015; Martins et al., 2017) focused only on the transformation of the knapsack formulation coefficients to ensure

the selection of best projects for the portfolio. Other studies proposed different objectives in order to i) maximize the overall efficiency of the portfolio (Liesiö et al., 2020) and ii) maximize the number of projects attaining given satisfaction levels on the criteria (Barbati et al., 2018). Recently, Kandakoglu et al. (2022) focused on maximizing the similarity of the selected projects for the portfolio. As a further study, developing alternate objective formulations can be explored in order to increase range of real-life PPS applications.

Multicriteria sorting/clustering for PPS problems: In recent years, although increasing attention has been placed on the development and application of multicriteria sorting and clustering methods (Alvarez et al., 2021; Amor et al., 2022), there are currently only two studies that employ sorting/clustering for a PPS problem. Mavrotas et al. (2008) first calculates augmented scores from the multicriteria sorting results, and then utilizes those in an optimization model with the relevant constraints to find the optimal portfolio. Kandakoglu et al. (2022) propose a novel robust multicriteria constrained clustering methodology that enables to group the similar projects into an optimal portfolio cluster that complies with the given constraints. For this purpose, the authors combine a novel Integer Programming (IP) model as an extension of the K-medoids clustering technique with the PROMETHEE method. Then, they enhance this combined model by embedding it into a SMAA framework to consider the uncertainties in multiple criteria evaluations, decision makers' preferences and real-world constraints. With these exceptions, to the best of our knowledge, multicriteria sorting and clustering have been largely ignored in the PPS field.

Here, we believe that there is a significant potential in using sorting or clustering methods for PPS problems. In particular, there seems to be two ways of benefiting from these methods. First, when the number of projects is relatively high (e.g., more than 100), the multicriteria sorting/clustering can be a well-suited option to reduce the size of the set of candidate projects. This may lead us to result in excluding the groups/clusters including the projects with low scores from further steps. By doing so, a more meaningful optimal portfolio with a better computational performance of the optimization model can be achieved. Second, in the case of having an objective of generating a portfolio of best projects in terms of high multicriteria evaluations, the PPS problem can be formulated as a constrained multicriteria sorting/clustering model to ensure the selection of best projects for the portfolio (Kandakoglu et al., 2022). In short, advances in sorting and clustering as well as machine learning provide opportunities and thus can be a promising future research area.

Robustness assessment: Although MCDA methods bring additional uncertainties (i.e., criteria weights, thresholds) into the PPS problem that should be considered in a robust setting, most of the studies focus on uncertainties in evaluations and preferences in PPS problems. From a theoretical measurement point only Liesiö et al. (2007, 2008), Mavrotas and Pechak (2013b), Mavrotas and Makryvelios (2021), Tervonen et al. (2017) and Kandakoglu et al. (2022) deal with a robustness assessment in portfolio evaluation.

Almost all of these studies addressed uncertainties in evaluations and preferences as well as resource constraints by using simulation-based approaches with different sets of parameters. However, the focus of the robustness assessment is the individual projects instead of portfolios. Hence, the need for robustness assessment for a portfolio remains a research challenge. Further studies should investigate other measures to assess the robustness of portfolios and to offer appropriate frameworks.

6 Conclusion

The use of MCDA methods for addressing the PPS problem has received increased attention in recent years, as more researchers and practitioners have demonstrated the appropriateness of these methods for dealing with multiple, conflicting and incommensurable criteria. These methods have also been proven to be suitable tools for complex PPS problems characterized by the existence of subjective evaluation processes, uncertainties and the participation of multiple stakeholders along with other characteristics.

This review has surveyed literature on discrete MCDA methods used to support the PPS process in order to analyze existing research and highlight research avenues for future work. After identifying the most representative articles in the literature, a classification of these articles that focuses on alternative approaches to determining a best or an optimal portfolio is included. Therefore, the individual project and portfolio evaluation problems are differentiated. The results highlight that this research field is highly heterogeneous in terms of its different areas of focus and procedures. Most studies utilize two-phased combined procedures and are mostly related to the individual project evaluation approach. Moreover, detailed information on how different MCDA methods are utilized with mathematical programming techniques in a combined manner is provided. For this purpose, the characteristics and formulations in each article are analyzed in details. In addition, we provided a summary of the reviewed papers as a decision tree as well as a rule set so that it can be effectively used as a decision support tool by scholars and practitioners for choosing an approach and method for a specific real-case PPS problem in hand.

We believe that understanding both the objective of the PPS decisions and the mathematical structure of the solutions can provide better decision aspects for various stakeholders. In this regard, the results of this review provide valuable information on existing literature for researchers, professionals and decision-makers dealing with real decision-making situations in a PPS process. Nevertheless, important research questions to be investigated still remain. As we have identified and discussed in detail in the previous section, there is a need for (1) investigating new formulations/methodologies to eliminate the commonly observed aforementioned drawbacks of the combined utilization of MCDA with optimization techniques, (2) extending the objective of portfolio decisions in order to increase the range of PPS applications, (3) using multicriteria sorting and clustering methods, and (4) conducting portfolio-level robustness assessment.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

Ahriz, S., El Yamami, A., Mansouri, K., & Qbadou, M. (2018). Cobit 5-based approach for it project portfolio management: Application to a moroccan university. *IJACSA: International Journal of Advanced Computer Science and Applications, 9*(4), 88–95.

Alvarez, P. A., Ishizaka, A., & Martínez, L. (2021). Multiple-criteria decision-making sorting methods: A survey. *Expert Systems with Applications, 183*, 115368.

Amor, S. B., Belaid, F., Benkraiem, R., Ramdani, B., & Guesmi, K. (2022). Multi-criteria classification, sorting, and clustering: A bibliometric review and research agenda. *Annals of Operations Research*. https://doi.org/10.1007/s10479-022-04986-9

Aragonés-Beltrán, P., Chaparro-González, F., Pastor-Ferrando, J.-P., & Pla-Rubio, A. (2014). An AHP (Analytic Hierarchy Process)/ANP (Analytic Network Process)-based multi-criteria decision approach for the selection of solar-thermal power plant investment projects. *Energy, 66*, 222–238.

Archer, N., & Ghasemzadeh, F. (2004). Project portfolio selection and management. *The Wiley Guide to Managing Projects*, pp. 237–255.

Archer, N., & Ghasemzadeh, F. (2007). Project portfolio selection and management. *Morris, P./Pinto, JK (2007), The Wiley Guide to Project, Program & Portfolio Management*, pp. 94–112.

Baker, E., Bosetti, V., & Salo, A. (2020). Robust portfolio decision analysis: An application to the energy research and development portfolio problem. *European Journal of Operational Research*.

Barbati, M., Greco, S., Kadziński, M., & Słowiński, R. (2018). Optimization of multiple satisfaction levels in portfolio decision analysis. *Omega, 78*, 192–204.

Bhattacharyya, R. (2015). A grey theory based multiple attribute approach for R& D project portfolio selection. *Fuzzy Information and Engineering, 7*(2), 211–225.

Carazo, A. F., Contreras, I., Gómez, T., & Pérez, F. (2012). A project portfolio selection problem in a group decision-making context. *Journal of Industrial & Management Optimization, 8*(1), 243–261.

Castillo, L., & Dorao, C. (2013). Decision-making in the oil and gas projects based on game theory: Conceptual process design. *Energy Conversion and Management, 66*, 48–55.

Chang, P.-T., & Lee, J.-H. (2012). A fuzzy DEA and knapsack formulation integrated model for project selection. *Computers & Operations Research, 39*(1), 112–125.

Cinelli, M., Kadziński, M., Gonzalez, M., & Słowiński, R. (2020). How to support the application of multiple criteria decision analysis? let us start with a comprehensive taxonomy. *Omega, 96*, 102261.

Clemen, R. T., & Smith, J. E. (2009). On the choice of baselines in multiattribute portfolio analysis: A cautionary note. *Decision Analysis, 6*(4), 256–262.

Danesh, D., Ryan, M. J., & Abbasi, A. (2018). Multi-criteria decision-making methods for project portfolio management: A literature review. *International Journal of Management and Decision Making, 17*(1), 75–94.

de Almeida, A. T., & Duarte, M. D. (2011). A multi-criteria decision model for selecting project portfolio with consideration being given to a new concept for synergies. *Pesquisa Operacional, 31*, 301–318.

de Almeida, A. T., & Vetschera, R. (2012). A note on scale transformations in the PROMETHEE V method. *European Journal of Operational Research, 219*(1), 198–200.

de Almeida, A. T., Vetschera, R., & de Almeida, J. A. (2014a). Scaling issues in additive multicriteria portfolio analysis. In *Decision Support Systems III-Impact of Decision Support Systems for Global Environments*, pp. 131–140. Springer.

de Almeida, J. A., de Almeida, A. T., & Costa, A. P. (2014). Portfolio selection of information systems projects using PROMETHEE V with c-optimal concept. *Pesquisa Operacional, 34*(2), 275–299.

De Souza, D. G. B., dos Santos, E. A., Soma, N. Y., & da Silva, C. E. S. (2021). MCDM-Based R&D Project Selection: A systematic literature review. *Sustainability, 13*(21), 11626.

Debnath, A., Roy, J., Kar, S., Zavadskas, E. K., & Antucheviciene, J. (2017). A hybrid MCDM approach for strategic project portfolio selection of agro by-products. *Sustainability, 9*(8), 1302.

Delouyi, F. L., & Ashrafi, M. (2021). Dynamic portfolio selection in gas transmission projects considering sustainable strategic alignment and project interdependencies through value analysis. *Sustainability, 13*(10), 5584.

Demircan Keskin, F. (2020). A two-stage fuzzy approach for industry 4.0 project portfolio selection within criteria and project interdependencies context. *Journal of Multi-Criteria Decision Analysis, 27*(1–2), 65–83.

Dixit, V., & Tiwari, M. K. (2020). Project portfolio selection and scheduling optimization based on risk measure: A conditional value at risk approach. *Annals of Operations Research, 285*(1–2), 9–33.

Doerner, K., Gutjahr, W. J., Hartl, R. F., Strauss, C., & Stummer, C. (2004). Pareto ant colony optimization: A metaheuristic approach to multiobjective portfolio selection. *Annals of Operations Research, 131*(1–4), 79–99.

Eilat, H., Golany, B., & Shtub, A. (2006). Constructing and evaluating balanced portfolios of R&D projects with interactions: A DEA based methodology. *European Journal of Operational Research, 172*(3), 1018–1039.

Fernandez, E., Gomez, C., Rivera, G., & Cruz-Reyes, L. (2015). Hybrid metaheuristic approach for handling many objectives and decisions on partial support in project portfolio optimisation. *Information Sciences, 315*, 102–122.

Frej, E. A., Ekel, P., & de Almeida, A. T. (2021). A benefit-to-cost ratio based approach for portfolio selection under multiple criteria with incomplete preference information. *Information Sciences, 545*, 487–498.

Frey, T., Buxmann, P., et al. (2012). It project portfolio management-a structured literature review. In *ECIS*, p. 167.

Ghapanchi, A. H., Tavana, M., Khakbaz, M. H., & Low, G. (2012). A methodology for selecting portfolios of projects with interactions and under uncertainty. *International Journal of Project Management, 30*(7), 791–803.

Guangul, F. M., Mebrahtu, G., & Chandra, S. (2021). Selection of Investment Portfolio Using Multi-criteria Decision Model (MCDM) and Integer Programming Model. In *Advances in Engineering Design*, pp. 101–116. Springer.

Hajiagha, S. H. R., Dahooie, J. H., Kandi, N. A., Zavadskas, E. K., & Xu, Z. (2022). Sustainable process selection using a hybrid Fuzzy DEMATEL and Fuzzy inference system. *International Journal of Fuzzy Systems*, pp. 1–18. https://doi.org/10.1007/s40815-021-01221-8.

Hashemizadeh, A., & Ju, Y. (2019). Project portfolio selection for construction contractors by MCDM–GIS approach. In *International Journal of Environmental Science and Technology*, pp 1–14.

Iamratanakul, S., Patanakul, P., & Milosevic, D. (2008). Project portfolio selection: From past to present. In *Management of Innovation and Technology, 2008. ICMIT 2008. 4th IEEE International Conference on IEEE*. pp. 287–292.

Jafarzadeh, M., Tareghian, H. R., Rahbarnia, F., & Ghanbari, R. (2015). Optimal selection of project portfolios using reinvestment strategy within a flexible time horizon. *European Journal of Operational Research, 243*(2), 658–664.

Jeng, D.J.-F., & Huang, K.-H. (2015). Strategic project portfolio selection for national research institutes. *Journal of Business Research, 68*(11), 2305–2311.

Kandakoglu, M., Walther, G., & Amor, S. B. (2022). A robust multicriteria clustering methodology for portfolio decision analysis. *Computers & Industrial Engineering, 174*, 108803.

Killen, C. P., Hunt, R. A., & Kleinschmidt, E. J. (2007). Managing the new product development project portfolio: a review of the literature and empirical evidence. In *Management of Engineering and Technology, Portland International Center for IEEE.*, pp. 1864–1874.

Kurttila, M., Haara, A., Juutinen, A., Karhu, J., Ojanen, P., Pykäläinen, J., Saarimaa, M., Tarvainen, O., Sarkkola, S., & Tolvanen, A. (2020). Applying a multi-criteria project portfolio tool in selecting energy peat production areas. *Sustainability, 12*(5), 1705.

Lee, J. W., & Kim, S. H. (2001). An integrated approach for interdependent information system project selection. *International Journal of Project Management, 19*(2), 111–118.

Liesiö, J., & Punkka, A. (2014). Baseline value specification and sensitivity analysis in multiattribute project portfolio selection. *European Journal of Operational Research, 237*(3), 946–956.

Liesiö, J., Mild, P., & Salo, A. (2007). Preference programming for robust portfolio modeling and project selection. *European Journal of Operational Research, 181*(3), 1488–1505.

Liesiö, J., Mild, P., & Salo, A. (2008). Robust portfolio modeling with incomplete cost information and project interdependencies. *European Journal of Operational Research, 190*(3), 679–695.

Liesiö, J., Andelmin, J., & Salo, A. (2020). Efficient allocation of resources to a portfolio of decision making units. *European Journal of Operational Research, 286*(2), 619–636.

Liesiö, J., Salo, A., Keisler, J. M., & Morton, A. (2021). Portfolio decision analysis: Recent developments and future prospects. *European Journal of Operational Research, 293*(3), 811–825. https://doi.org/10.1016/j.ejor.2020.12.015

Lopes, Y. G., & de Almeida, A. T. (2015). Assessment of synergies for selecting a project portfolio in the petroleum industry based on a multi-attribute utility function. *Journal of Petroleum Science and Engineering, 126*, 131–140.

Ma, J., Harstvedt, J. D., Jaradat, R., & Smith, B. (2020). Sustainability driven multi-criteria project portfolio selection under uncertain decision-making environment. *Computers & Industrial Engineering, 140*, 106236.

Mahmoudi, A., Abbasi, M., & Deng, X. (2022). A novel project portfolio selection framework towards organizational resilience: robust ordinal priority approach. *Expert Systems with Applications, 188*, 116067.

Marques, A. C., Frej, E. A., & de Almeida, A. T. (2022). Multicriteria decision support for project portfolio selection with the fitradeoff method. *Omega, 111*, 102661.

Martins, C. L., López, H. M. L., de Almeida, A. T., Almeida, J. A., & Bortoluzzi, M. B. (2017). An MCDM project portfolio web-based DSS for sustainable strategic decision making in an electricity company. *Industrial Management & Data Systems, 117*(7), 1362–1375.

Martinsuo, M., & Geraldi, J. (2020). Management of project portfolios: Relationships of project portfolios with their contexts. *International Journal of Project Management*. https://doi.org/10.1016/j.ijproman.2020.02.002

Mashayekhi, Z., & Omrani, H. (2016). An integrated multi-objective markowitz-dea cross-efficiency model with fuzzy returns for portfolio selection problem. *Applied Soft Computing, 38*, 1–9.

Mavrotas, G., & Makryvelios, E. (2021). Combining multiple criteria analysis, mathematical programming and monte carlo simulation to tackle uncertainty in research and development project portfolio selection: A case study from greece. *European Journal of Operational Research, 291*(2), 794–806.

Mavrotas, G., & Pechak, O. (2013a). Combining mathematical programming and Monte Carlo simulation to deal with uncertainty in energy project portfolio selection. In *Assessment and Simulation Tools for Sustainable Energy Systems*, pp. 333–356. Springer.

Mavrotas, G., & Pechak, O. (2013). The trichotomic approach for dealing with uncertainty in project portfolio selection: Combining MCDA, mathematical programming and Monte Carlo simulation. *International Journal of Multicriteria Decision Making, 3*(1), 79–96.

Mavrotas, G., Diakoulaki, D., & Capros, P. (2003). Combined MCDA-IP approach for project selection in the electricity market. *Annals of Operations Research, 120*(1–4), 159–170.

Mavrotas, G., Diakoulaki, D., & Caloghirou, Y. (2006). Project prioritization under policy restrictions. A combination of MCDA with 0–1 programming. *European Journal of Operational Research, 171*(1), 296–308.

Mavrotas, G., Diakoulaki, D., & Kourentzis, A. (2008). Selection among ranked projects under segmentation, policy and logical constraints. *European Journal of Operational Research, 187*(1), 177–192.

Meskendahl, S. (2010). The influence of business strategy on project portfolio management and its success-a conceptual framework. *International Journal of Project Management, 28*(8), 807–817.

Mild, P., Liesiö, J., & Salo, A. (2015). Selecting infrastructure maintenance projects with robust portfolio modeling. *Decision Support Systems, 77*, 21–30.

Morton, A. (2015). Measurement issues in the evaluation of projects in a project portfolio. *European Journal of Operational Research, 245*(3), 789–796.

Morton, A., Keisler, J. M., & Salo, A. (2016). Multicriteria portfolio decision analysis for project selection. In *Multiple Criteria Decision Analysis*, pp. 1269–1298. Springer.

Müller, R., Martinsuo, M., & Blomquist, T. (2008). Project portfolio control and portfolio management performance in different contexts. *Project Management Journal, 39*(3), 28–42.

Mussoi, F. L. R., & Teive, R. C. G. (2021). An integrated multicriteria decision-making approach for distribution system expansion planning. *International Journal of Intelligent Systems, 36*(9), 4962–4989.

Nowak, M., & Trzaskalik, T. (2022). A trade-off multiobjective dynamic programming procedure and its application to project portfolio selection. *Annals of Operations Research, 311*, 1155–1181.

Parreiras, R., Kokshenev, I., Carvalho, M., Willer, A., Dellezzopolles, C., Jr., Nacif, D., Jr., & Santana, J. (2019). A flexible multicriteria decision-making methodology to support the strategic management of science, technology and innovation research funding programs. *European Journal of Operational Research, 272*(2), 725–739.

Parvaneh, F., & El-Sayegh, S. M. (2016). Project selection using the combined approach of AHP and LP. *Journal of Financial Management of Property and Construction, 21*(1), 39–53.

Petchrompo, S., Coit, D. W., Brintrup, A., Wannakrairot, A., & Parlikad, A. K. (2022). A review of pareto pruning methods for multi-objective optimization. *Computers & Industrial Engineering, 167*, 108022.

Petchrompo, S., Wannakrairot, A., & Parlikad, A. K. (2022). Pruning pareto optimal solutions for multi-objective portfolio asset management. *European Journal of Operational Research, 297*(1), 203–220.

Raad, N., Shirazi, M., & Ghodsypour, S. (2020). Selecting a portfolio of projects considering both optimization and balance of sub-portfolios. *Journal of Project Management, 5*(1), 1–16.

Raad, N. G., & Shirazi, M. A. (2020). A project portfolio selection framework for transforming Iranian universities into entrepreneurial institutions. *Journal of Industry-University Collaboration*. https://doi.org/10.1108/JIUC-06-2019-0014

Rad, F. H., & Rowzan, S. M. (2018). Designing a hybrid system dynamic model for analyzing the impact of strategic alignment on project portfolio selection. *Simulation Modelling Practice and Theory, 89*, 175–194.

Ranjbar, M., Nasiri, M. M., & Torabi, S. A. (2022). Multi-mode project portfolio selection and scheduling in a build-operate-transfer environment. *Expert Systems with Applications, 189*, 116134.

Relich, M., & Pawlewski, P. (2017). A fuzzy weighted average approach for selecting portfolio of new product development projects. *Neurocomputing, 231*, 19–27.

Rudnik, K., Bocewicz, G., Kucińska-Landwójtowicz, A., & Czabak-Górska, I. D. (2021). Ordered fuzzy WASPAS method for selection of improvement projects. *Expert Systems With Applications, 169*, 114471.

Salo, A., Keisler, J., & Morton, A. (2011). Portfolio decision analysis: improved methods for resource allocation, vol. 162. Springer. https://doi.org/10.1007/978-1-4419-9943-6_1

Shakhsi-Niaei, M., Torabi, S. A., & Iranmanesh, S. H. (2011). A comprehensive framework for project selection problem under uncertainty and real-world constraints. *Computers & Industrial Engineering, 61*(1), 226–237.

Sharifighazvini, M., Ghezavati, V., Raissi, S., & Makui, A. (2018). Integration of a new MCDM approach based on the DEA, FANP with MONLP for efficiency-risk assessment to optimize project portfolio by branch and bound: A real case-study. *Economic Computation and Economic Cybernetics Studies and Research, 52*(1), 261–278.

Song, S., Yang, F., & Xia, Q. (2019). Multi-criteria project portfolio selection and scheduling problem based on acceptability analysis. *Computers & Industrial Engineering, 135*, 793–799.

Song, S., Wei, T., Yang, F., & Xia, Q. (2021). Stochastic multi-attribute acceptability analysis-based heuristic algorithms for multi-attribute project portfolio selection and scheduling problem. *Journal of the Operational Research Society, 72*(6), 1373–1389.

Szilágyi, I., Sebestyén, Z., & Tóth, T. (2020). Project ranking in petroleum exploration. *The Engineering Economist, 65*(1), 66–87.

Tavana, M., Khalili-Damghani, K., & Abtahi, A.-R. (2013). A fuzzy multidimensional multiple-choice knapsack model for project portfolio selection using an evolutionary algorithm. *Annals of Operations Research, 206*(1), 449–483.

Tavana, M., Keramatpour, M., Santos-Arteaga, F. J., & Ghorbaniane, E. (2015). A fuzzy hybrid project portfolio selection method using data envelopment analysis, TOPSIS and integer programming. *Expert Systems with Applications, 42*(22), 8432–8444.

Tavana, M., Khosrojerdi, G., Mina, H., & Rahman, A. (2019). A hybrid mathematical programming model for optimal project portfolio selection using fuzzy inference system and analytic hierarchy process. *Evaluation and Program Planning, 77*, 101703.

Tavana, M., Khosrojerdi, G., Mina, H., & Rahman, A. (2020). A new dynamic two-stage mathematical programming model under uncertainty for project evaluation and selection. *Computers & Industrial Engineering, 149*, 106795.

Teller, J., & Kock, A. (2013). An empirical investigation on how portfolio risk management influences project portfolio success. *International Journal of Project Management, 31*(6), 817–829.

Tervonen, T., Liesiö, J., & Salo, A. (2017). Modeling project preferences in multiattribute portfolio decision analysis. *European Journal of Operational Research, 263*(1), 225–239.

Urli, B., & Terrien, F. (2010). Project portfolio selection model, a realistic approach. *International Transactions in Operational Research, 17*(6), 809–826.

Vetschera, R., & de Almeida, A. T. (2012). A PROMETHEE-based approach to portfolio selection problems. *Computers & Operations Research, 39*(5), 1010–1020.

Wu, Y., Xu, C., Ke, Y., Chen, K., & Sun, X. (2018). An intuitionistic fuzzy multi-criteria framework for large-scale rooftop pv project portfolio selection: Case study in zhejiang, china. *Energy, 143*, 295–309.

Wu, Y., Xu, C., Ke, Y., Li, X., & Li, L. (2019). Portfolio selection of distributed energy generation projects considering uncertainty and project interaction under different enterprise strategic scenarios. *Applied Energy, 236*, 444–464.

Wu, Y., Xu, C., Ke, Y., Tao, Y., & Li, X. (2019). Portfolio optimization of renewable energy projects under type-2 fuzzy environment with sustainability perspective. *Computers & Industrial Engineering, 133*, 69–82.

Yang, F., Song, S., Huang, W., & Xia, Q. (2015). SMAA-PO: Project portfolio optimization problems based on stochastic multicriteria acceptability analysis. *Annals of Operations Research, 233*(1), 535–547.

Yu, L., Wang, S., Wen, F., & Lai, K. K. (2012). Genetic algorithm-based multi-criteria project portfolio selection. *Annals of Operations Research, 197*(1), 71–86.

Zandi, F., & Tavana, M. (2010). A multi-attribute group decision support system for information technology project selection. *International Journal of Business Information Systems, 6*(2), 179–199.

Zhang, X., Hipel, K. W., & Tan, Y. (2019). Project portfolio selection and scheduling under a fuzzy environment. *Memetic Computing, 11*(4), 391–406.

Zhang, X., Fang, L., Hipel, K. W., Ding, S., & Tan, Y. (2020). A hybrid project portfolio selection procedure with historical performance consideration. *Expert Systems with Applications, 142*, 113003.

Zhu, Q., Dou, Y., & Sarkis, J. (2010). A portfolio-based analysis for green supplier management using the analytical network process. *Supply Chain Management: An International Journal, 15*(4), 306–319.

Zorluoglu, Ö. S., & Kabak, Ö. (2021). A literature survey on project portfolio selection problem. In *Multiple criteria decision making: Beyond the information age*, Springer. pp. 387–411.

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